Personas, Jobs-to-be-Done & the RBAC Role Model

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Revision: 3 **Last modified:** 2026-07-04T12:00:00Z

> **Rev 3 (2026-07-04, independent gap-analysis pass).** §2.2 (business-admin / tenant-owner) JTBD extended with the MSP-operator's incident-response and compliance-reporting concerns — an operator running networks for several client tenants needs an audit trail it can *export per-incident* and *reason about across tenants it administers*, not only view live in Console. No other change; the four-axis model, RBAC enum, and persona↔role mapping are unchanged and remain reconciled to `svc-identity.md`.

> **Reconciled (§11.4.35, 2026-06-26):** the RBAC role set is fixed to the authoritative `svc-identity.md` enum `{member, operator, admin}` (+ **Device**, which holds **no RBAC** and is policy-gated). The earlier invented **"tenant owner"** RBAC role is **removed** — `admin` is the top RBAC role — and the missing **`operator`** role (mint enroll tokens / manage devices + policies + advertise prefixes) is **added**. "Self-hoster / tenant-owner" is modelled as a **host/deployment privilege**, NOT an RBAC role. The connector-operator's `route.advertised` capability is **`operator`-tier**, not `member`. This aligns the document to `use-cases-and-journeys.md` UC-04 (admin/operator) and the `svc-identity.md` §7.3 `minRole` table, which win on any conflict.

> **Document role.** Volume 1 deep document that expands §5 (personas and the three app classes) and §3 (the three system roles) of the overview [`00-product-scope-and-principles.md`]() into a full persona catalogue (jobs-to-be-done, journeys, pain points, success criteria) **and** the RBAC role/

> capability model. It distinguishes two orthogonal axes that the overview keeps separate and that downstream readers routinely conflate:
>
> 1. **System roles** — *Connector / Gateway / Client* — the topological role a node plays in the data path [00 §3], [SPEC §3].
> 2. **Identity/RBAC roles** — *admin / operator / member / device (anon-device-token)* — the authorization a principal holds in the control plane [00 §12 "Tenant"], (security overview S1–S5), per `svc-identity.md`.
>
> A persona is a *human*; an app class is the *software* a persona drives; a system role is the *topological position*; an RBAC role is the *permission grant*. This document maps all four.
>
> **Status:** SPEC-ONLY. The RBAC capability matrix here is the *product-level* view; the authoritative API-level permission contract is `../v03-control-plane/svc-identity.md` and the enforcement points are `../04-security-privacy-pki.md` (S1/S3). Where this document and svc-identity disagree on a capability, svc-identity wins (§11.4.35). Any capability not yet fixed by svc-identity is marked `UNVERIFIED`.
>
> **Evidence base.** `[00 §N]` = the overview; `[SPEC §N]` = the spine; `[SEC S#]` = the security overview invariants `../04-security-privacy-pki.md`.

Table of contents

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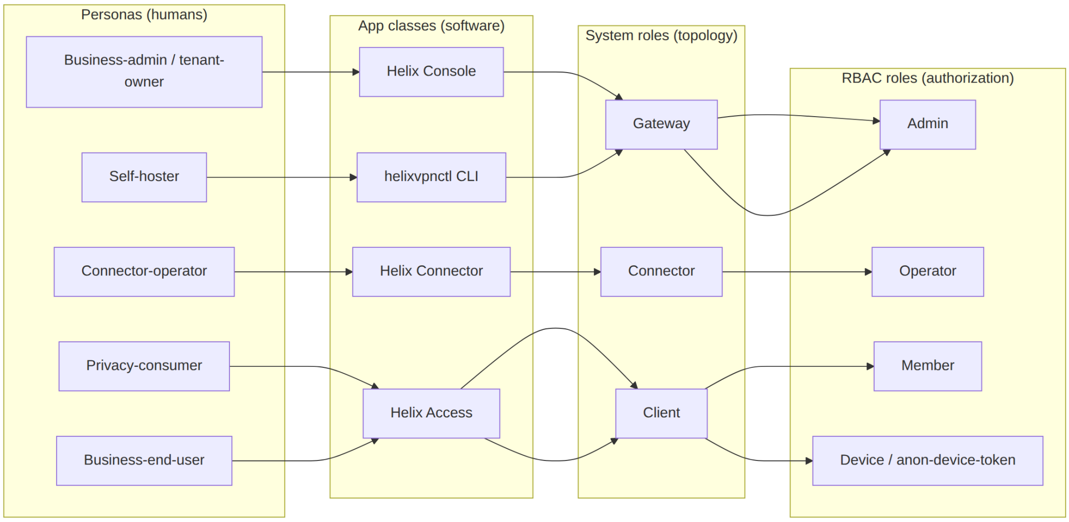
1. The four-axis model



The mapping is *not* one-to-one: a self-hoster holds the `admin` RBAC role *and* the host/deployment-owner privilege *and* operates Connectors *and* uses an Access client — one human, several roles. ("Tenant-ownership / self-hosting" is a host/deployment privilege, not an RBAC role — see §4.) The matrix in §5 makes every cell explicit so no capability is assumed.

Axis	Closed set	Source of truth
Persona	privacy-consumer, business-admin/tenant-owner, business-end-user, connector-operator, self-hoster	this doc §2
App class	Helix Access, Helix Console, Helix Connector, `helixvpnctl`	[00 §5], [SPEC §3]
System role	Client, Gateway, Connector	[00 §3], [SPEC §3]
RBAC role	admin, operator, member, device (anon-device-token)	[00 §12], [SEC S1–S5], svc-identity (authoritative enum)

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2. Persona catalogue

Each persona is given: **JTBD** (jobs-to-be-done), **primary journey**, **pain points it removes**, and **success criteria** (tied where possible to the 8-criteria MVP Definition-of-Done [00 §10.2], [SPEC §8.1]). Success criteria that are not yet a fixed acceptance gate are marked `UNVERIFIED`.

2.1 Privacy-consumer

> "*I want into a privacy exit (or my own lab) with one button, no email, and no way for anyone to log what I do.*" — the Mullvad use case [00 §5.1].

Field	Detail
Drives	Helix Access app (iOS, Android, desktop; Web limited)
System role	Client [00 §3.3]
RBAC role	Device, via **anonymous device token** (no email) [00 §9 F15]
JTBD	(1) connect to a privacy exit with one tap; (2) get automatic obfuscation that just works when a network blocks WG; (3) never leak when the tunnel drops; (4) hold an anonymous identity with no PII
Primary journey	install → anonymous device-token enrollment (key generated on-device, private key never leaves) → one-button connect → auto-ladder escalates plain → LWO → MASQUE if blocked → kill-switch + DNS-leak protection active throughout [00 §9 F7/F11/F13/F15], [SEC S2/S9]
Pain points removed	account email requirement (anonymous token); manual obfuscation fiddling (auto-ladder); leak-on-drop (core-owned kill-switch); trust-the-vendor no-logs (no-logging-as-code)
Success criteria	DoD#4 (auto-escalate to MASQUE when WG blocked); DoD#7 (kill-switch + DNS-leak verified, no leak on drop); anonymous enrollment succeeds with no PII captured
Sensitivities	memory- and battery-sensitive on mobile [00 §5.1]; the iOS NE ~15 MB ceiling is a hard constraint (G3)

2.2 Business-admin / tenant-owner

> "*I manage tenants, users, devices and policy for an organization (or several client orgs), and I need a live topology view and an audit trail of control actions.*" [00 §5.3]

Field	Detail
Drives	Helix Console (Web responsive + Desktop — same Flutter build, **no tunnel core**) [00 §5, §5.2]
System role	

Field	Detail
	none in the data path — Console is API-client-only, drives the Gateway control plane [00 §5]
RBAC role	**Admin** (the top RBAC role; tenant-ownership is a host/deployment privilege, not an RBAC role)
JTBD	(1) CRUD tenants/users/devices/networks/routes/policies; (2) see/revoke devices instantly; (3) author default-deny ACLs that compile to reachability; (4) read a control-action audit trail; (5) optionally manage multi-tenant billing; (6) **for an MSP administering several client tenants:** export an audit slice for a single incident/tenant on demand (compliance reporting to that client), and run a first-response revoke/isolate action within the same <1 s enforcement guarantee as any other device revoke
Primary journey	log in (OIDC) → create a tenant → invite members → approve/enroll devices → author policy (`group:contractors → net:warehouse:554/tcp`) → watch it converge in <1 s → revoke a lost device and watch enforcement in <1 s → review `audit_events` [00 §9 F17], [SPEC §8.1 DoD#5/#6], [SEC S5/S7]
Incident-response / compliance journey (MSP-operator variant)	suspected-compromised-device report arrives → admin locates the device in Console (own-tenant scope, RLS-bounded) → revokes it (<1 s edge enforcement, DoD#6) → filters `audit_events` to that device/time-window → exports the filtered control-action slice as the incident evidence handed to the affected client — never a traffic/destination record, because none exists (P7) [SEC S7], `audit-and-compliance.md`
Pain points removed	policy edits requiring restarts (push-based, <1 s); device revoke lag (<1 s edge enforcement); destinations/flows leaking into audit (audit is control-only); an MSP having to promise "trust me" on incident evidence instead of handing over an exportable control-action trail scoped to the affected tenant only
Success criteria	DoD#5 (policy edit reflected <1 s, no restart); DoD#6 (device revoke enforced <1 s); audit records who-did-what-to-identity/policy/devices, never destinations [SEC S7]; an audit export for one tenant/time-window

Field	Detail
	contains **zero** rows from any other tenant (RLS-bounded export, not just RLS-bounded live view)
Sensitivities	tenant isolation must be airtight — Postgres RLS, not app-layer only [SEC S6/RLS]; an MSP-operator's cross-tenant *administrative* access (if any) is itself an audited action — administering tenant B is never silent to tenant B's own audit trail

2.3 Business-end-user

> "*My admin set me up; I just want into the work networks I'm allowed to reach, and to be cleanly denied the ones I'm not.*" (derived from the ZTNA half of the dual model [00 §3, §4.2])

Field	Detail
Drives	Helix Access app
System role	Client
RBAC role	**Member** (an identity that belongs to a tenant, holds enrolled devices, but cannot administer the tenant)
JTBD	(1) connect and reach exactly the subset of joined networks policy grants; (2) be transparently denied unauthorized networks (default-deny, not an error to debug); (3) see only the peers/routes need-to-know permits
Primary journey	receive an enrollment invite/token from the admin → device generates its WG keypair (private key never leaves) → enroll → receive a **policy-filtered** network map (only granted peers/routes) → reach authorized LAN host, denied unauthorized one [00 §4.2, §9 F17], [SEC S1/S3]
Pain points removed	over-broad access (need-to-know map — the user never even learns of peers it cannot reach [SEC S3]); confusing all-or-nothing VPNs (policy-scoped 1→N)
Success criteria	DoD#3 (reach an authorized LAN host AND deny an unauthorized one — default-deny proven); the user's map contains only granted peers [SEC S3]
Sensitivities	must not be able to enumerate the tenant topology beyond its grants (need-to-know is server-side, pre-wire)

2.4 Connector-operator

> "*Install one agent inside my LAN, it dials out, done — I never touch my router.*" [00 §5.2]

Field	Detail
Drives	Helix Connector (headless daemon on Linux/Windows/macOS + optional slim UI; Android/embedded for appliances) [00 §3.1, §5]
System role	**Connector** [00 §3.1]
RBAC role	**Operator** — the connector identity mints/redeems enroll tokens, manages devices + policies, and advertises prefixes; the `route.advertised` capability is **`operator`-tier per `svc-identity.md`**, not `member`. A self-hoster operating their own connector also holds **`admin`** (§2.5)
JTBD	(1) onboard a network without opening any inbound port; (2) advertise the CIDRs the network exposes (`route.advertised`); (3) set local ACLs; (4) run headless and reconnect resiliently
Primary journey	install agent on a LAN host → enroll (device cert + enrollment token) → dial outbound to the Gateway (reverse tunnel) → advertise CIDRs → Gateway routes authorized client traffic into the LAN and responses back out — neither side ever opened an inbound port [00 §3.1, §4.1, §8 P3]
Pain points removed	router port-forwarding (outbound-only); inbound attack surface (none); per-network bespoke VPN configs (same Rust core in advertise/route mode)
Success criteria	DoD#2 (enroll a Connector); DoD#3 (an authorized client reaches a LAN host behind the advertised CIDR); the Connector establishes the tunnel with zero inbound exposure
Sensitivities	overlapping RFC1918 ranges across connectors must not collide (D4 → 4via6) [00 §4.2 #1]

2.5 Self-hoster

> "*I run the whole thing myself — Gateway, Connector(s), and my own Access client — because no-logs is credible only when I own the box.*" [00 §7.1]

Field	Detail
Drives	

Field	Detail
	`helixvpnctl` (Cobra CLI) for bring-up + Console + Connector + Access — wears **all** software hats
System role	operates Gateway, one or more Connectors, and a Client
RBAC role	**Admin** of their own tenant(s) — the top RBAC role. **Tenant-ownership / self-hosting is a host/deployment privilege**, NOT an RBAC role: the human who owns the box bootstraps each tenant (`helixvpnctl init`) and then administers it as `admin`. A self-hoster running networks for multiple clients administers multiple tenants [00 §12]
JTBD	(1) stand up a Gateway from zero on a clean VPS with one command; (2) own the no-logs guarantee by owning the deployment; (3) scale the same images from one homelab pod to an HA fleet without a rewrite
Primary journey	`helixvpnctl init` on a clean VPS (one rootless Podman pod: edge + control + Postgres + Redis) → enroll a Connector and a Client → reach a LAN host → confirm no durable connection log exists (schema-lint + runtime check) [00 §7.1, §10.2 DoD#1/#8], [SPEC §8.1]
Pain points removed	SaaS coordination dependency (self-hosted); vendor-trusted no-logs (self-evident ownership + CI-enforced schema); separate codebases for homelab vs fleet (same images, P6)
Success criteria	DoD#1 (self-host from zero via `helixvpnctl init`); DoD#8 (no durable connection log AND all three apps drive the system); the same images scale to HA (Phase 2)
Sensitivities	the CA root key + Postgres are the only secrets to protect — data-plane nodes are cattle [SEC S11]

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3. Persona → app-class → system-role → RBAC-role mapping

Persona	Primary app class	Other apps used	System role(s)	RBAC role(s)	Uses tunnel core?
Privacy-consumer	Helix Access	—	Client	Device (anon-	yes (`helix-core`

Persona	Primary app class	Other apps used	System role(s)	RBAC role(s)	Uses tunnel core?
				device-token)	capture mode)
Business-admin / tenant-owner	Helix Console	(Access as a user)	none in data path	Admin	no (Console is API-client only)
Business-end-user	Helix Access	—	Client	Member (+ Device per enrolled device)	yes
Connector-operator	Helix Connector	(Console to verify routes)	Connector	Operator (+ Device); or Admin if self-hosting	yes (`helix-core` advertise/ route mode)
Self-hoster	`helixvpnctl` + Console + Connector + Access	all	Gateway + Connector + Client	Admin (+ host/ deployment-owner privilege)	yes (as Connector/ Client)

> **Note (the single-tree flavoring contract, [00 §5.2]).** All three apps build from one Flutter tree via `runHelixApp(flavor, home, capabilities)`, `flavor ∈ {Access, Connector, Console}`; **Console is the only build that omits `core_ffi`** (no Rust tunnel core). The "Uses tunnel core?" column above is a direct consequence of this contract. Owned by `../03-client-core-and-ui.md`.

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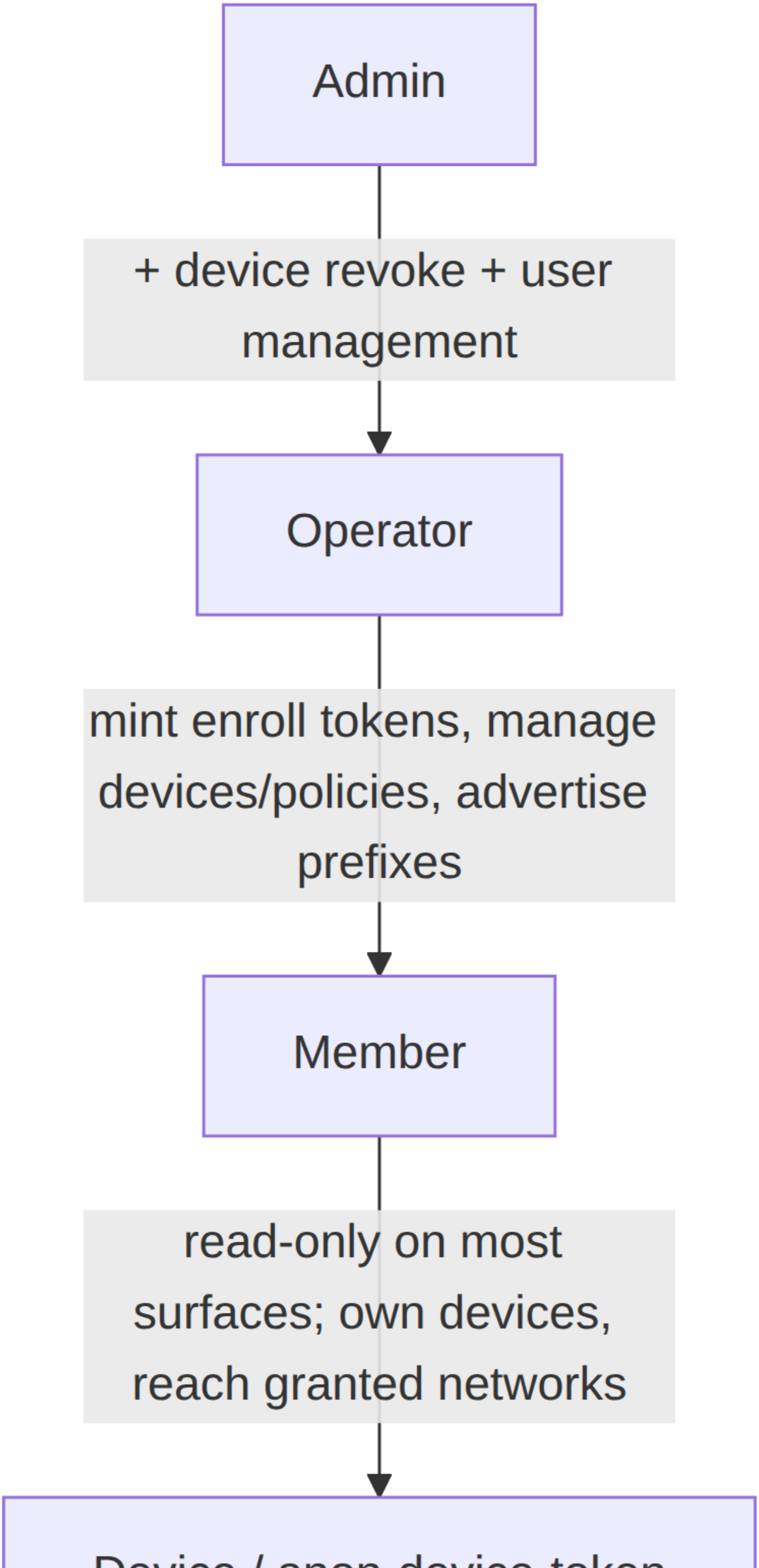
4. The RBAC role model

The control plane authorizes principals by **RBAC role within a tenant**. A **tenant** is an isolated organization boundary enforced by Postgres RLS [00 §12], [SEC S6/RLS]. The closed role set is the authoritative `svc-identity.md` enum `{member, operator, admin}` (+ Device, which holds **no RBAC** and is policy-gated). **Tenant-ownership / self-hosting is a host/deployment privilege, not a member of this RBAC enum.** The role lattice (`admin > operator > member`, per `svc-identity.md` §7.3):


Admin
+ device revoke + user management
Operator
mint enroll tokens, manage devices/policies, advertise prefixes
Member
read-only on most surfaces; own devices, reach granted networks

4.1 Role definitions

RBAC role	Definition	Distinguishing capability	Authenticated by
Admin	The **top** RBAC role within a tenant — everything an Operator can do, plus device revoke and user management; the business-admin's and self-hoster's role	device revoke; user management; full CRUD on users/ devices/ networks/routes/ policies	OIDC identity [00 §9 F15]
Operator	Manages a tenant's devices and policies and mints enrollment tokens; the connector-operator's role	mint/redeem enroll tokens; manage devices/ policies; **advertise prefixes (`route.advertised`)**	OIDC identity
Member	A user that belongs to a tenant and owns enrolled devices; **read-only on most administrative surfaces**	enroll own devices; connect; reach granted networks	OIDC identity
Device / anon-device-token	A *data-plane* principal — an enrolled device holding its own WG keypair; **holds no RBAC role (policy-gated, never the RBAC matrix)**; the anonymous-device-token variant has no human identity behind it (the	open `WatchNetworkMap`; reach `AllowedIPs`; nothing administrative	short-lived mTLS device cert (≤24 h, tenant-CA-signed) [SEC S4], + WG Noise IK for data [SEC §1.2]

RBAC role	Definition	Distinguishing capability	Authenticated by
	privacy-consumer case)		

> **Host/deployment privilege (not RBAC).** Tenant lifecycle (create/delete a tenant), bringing up the Gateway (`helixvpnctl init`), and any managed-SKU billing are **host/deployment-owner** capabilities the self-hoster exercises by owning the box — they are **not** entries in the `{member, operator, admin}` RBAC enum. `svc-identity.md` §7.3's `minRole` table (the authoritative `Permission → Role` map) tops out at `admin` (user management + device revoke); it has no "tenant owner" tier.

4.2 Two authentication channels, never conflated

Per the security overview [SEC §1.2], a principal authenticates on two *independent* channels:

1. **Control channel** (`Coordinator.WatchNetworkMap/.Enroll/…`): short-lived device cert (mTLS, ≤24 h, auto-renew), tenant-CA-signed [SEC S4].
2. **Data channel** (the tunnel): WireGuard Noise IK; peer pubkeys delivered via the policy-filtered map [SEC §1.2].

Compromising one does not yield the other. A device must be valid on **both** to *learn* its peers and *reach* them. This is why the **Device** RBAC role is the join point between identity (control) and reachability (data).

4.3 Default-deny is the RBAC ground state (S1)

A Member's device with no policy grant has an **empty `AllowedIPs` and no edge verdict entry** — cryptographically and administratively isolated even though enrolled and online [SEC S1, §1.3]. RBAC roles grant *administrative* capability; *reachability* is a separate, additive, default-empty grant compiled by the policy service. Holding the Member role does **not** imply reaching any network.

4.4 Anonymous device tokens (the privacy lane)

The anonymous-device-token enrollment is a first-class path alongside OIDC [00 §9 F15]: a device enrolls and obtains a control identity with **no email / no PII**, generating its WG keypair on-device (private key never leaves) [SEC S2]. This is the privacy-consumer's principal — a Device (holding no RBAC role) with no human member/operator/admin behind it. Owned by `../v05-security/identity-and-enrollment.md`

- `../v03-control-plane/svc-identity.md`.

> **UNVERIFIED** — the precise capability ceiling of an anonymous-device-token principal (e.g. whether it may advertise prefixes, or only consume a privacy-exit) is fixed by `svc-identity`, not here. Until svc-identity pins it, this document assumes anon tokens are **privacy-exit/consume-only** (the consumer case), which is the conservative default per §11.4.6.

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5. Persona × capability matrix

Capabilities are grouped by control-plane domain. Legend: allowed · denied · allowed only on own resources · 🅿 policy-gated (granted additively, default-deny §4.3). The authoritative permission contract is `svc-identity`; rows it has not yet fixed are `UNVERIFIED` (flagged inline).

Capability	Privacy-consumer (Device/ anon)	Business-end-user (Member)	Connector-operator (Operator)	Business-admin (Admin)	Se (A ho pr
Connect / open tunnel			(advertise/ route mode)	n/a (Console)	
Use Gateway as privacy exit			n/a	n/a	
Reach a joined private network	(anon = exit-only, `UNVERIFIED`)	🅿	🅿 (its own LAN)	n/a	🅿
Advertise CIDRs (`route.advertised`)			(for its connector)	(set on behalf)	
Enroll a device	(self only)	(own devices)	(own connector)	(any in tenant)	
See own devices				(all)	
Revoke a device	(own)	(own)	(own connector)	(any, <1 s)	
Author / edit ACL policy			(Operator tier = `PermManagePolicy`; local-vs-central scope `UNVERIFIED`)		

Capability	**Privacy-consumer (Device/ anon)**	**Business-end-user (Member)**	**Connector-operator (Operator)**	**Business-admin (Admin)**	**Se (A ho pr**
CRUD users					
CRUD networks / routes			(its own routes)		
Read control-action audit					
Manage tenants (create/delete)				(host/ deployment privilege, not RBAC; `UNVERIFIED` whether ever exposed to `admin`)	de pr no
Manage billing (optional SKU)				(host/ deployment / managed-SKU privilege; `UNVERIFIED`)	de ma SK pr
Bring up Gateway (`helixvpnctl init`)					de pr no

> **Honesty note (§11.4.6).** The matrix is the *product-intent* view. Three cells are `UNVERIFIED` because the exact grant is fixed by `svc-identity`/`svc-policy`, not by a product narrative: (a) whether an anon device token may reach a private network or is exit-only; (b) the local-vs-central scope of a Connector-operator's (Operator-tier) policy authoring; (c) whether tenant-lifecycle and billing — both **host/ deployment privileges, not RBAC** — are ever surfaced through the `admin` role or remain box-owner-exclusive. Each is a tracked refinement item; the conservative default (narrower grant) is assumed until pinned, per §11.4.6/ §11.4.101.

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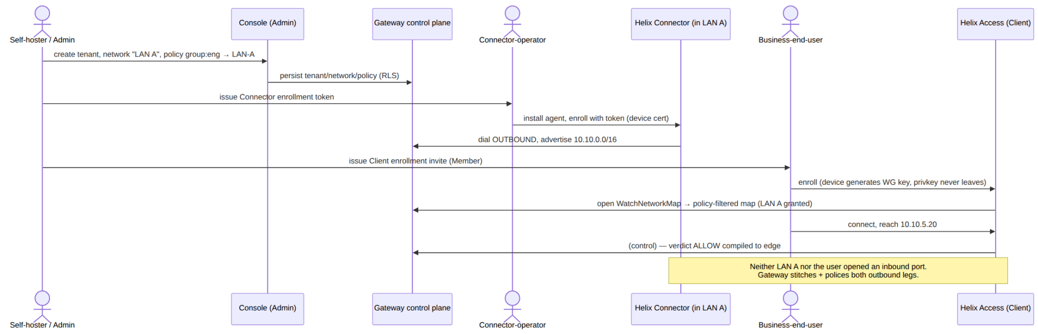
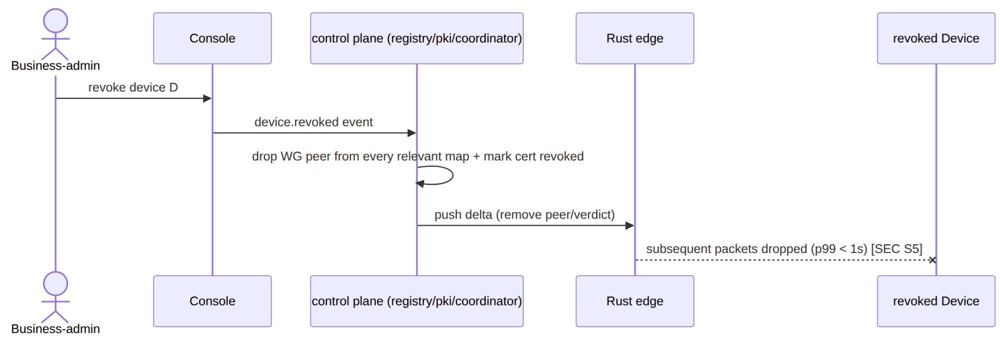
6. Cross-persona journeys

6.1 Onboarding a network so a remote user can reach it (the founding journey)

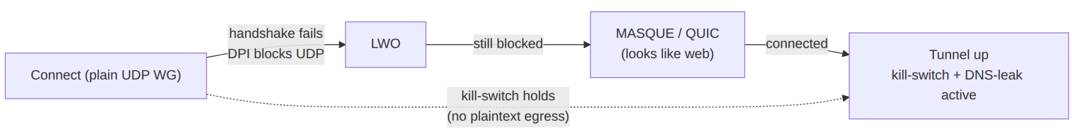
This is the canonical multi-persona flow that realizes the founding constraint [00 §2.1, §4.1] — three personas, no inbound port-forward:



6.2 Lost-device revocation (admin + device, <1 s)



6.3 Privacy-consumer auto-obfuscation under DPI (one persona, one journey)



The escalation ladder is driven by handshake-failure events; throughout, the kill-switch holds so no plaintext egress occurs while escalating [00 §9 F7/F11], [SEC S9].

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7. Cross-document contracts this document fixes

1. **Four-axis separation** (§1) — persona ≠ app class ≠ system role ≠ RBAC role; every later document MUST keep these distinct (the overview conflates persona and app class loosely in §5; this document is the disambiguation authority for Volume 1).
2. **Persona → RBAC mapping** (§3, §5) — consumed by `functional-requirements.md` (each FR names the persona/role it serves) and bound to the authoritative `../v03-control-plane/svc-identity.md`.
3. **Device is the join point** of control identity and data reachability (§4.2) — inherited from [SEC §1.2], not redefined.
4. **Default-deny is the RBAC ground state** (§4.3) — RBAC grants administrative capability; reachability is a separate additive default-empty grant ([SEC S1], `../v03-control-plane/svc-policy.md`).
5. **Anonymous device tokens are first-class** (§4.4) — the privacy lane; owned by `svc-identity` + `identity-and-enrollment.md`.
6. **The single-tree flavoring contract** (§3 note) governs "which persona's app carries the tunnel core" — inherited from [00 §5.2].

Sources verified

- `00-product-scope-and-principles.md` §3 (three roles), §4 (two-way + multi-network), §5 + §5.1/§5.2 (personas, app classes, flavoring contract), §9 (parity matrix — F15 anon account, F17 device mgmt), §10.2 (8-criteria DoD), §12 (glossary — tenant).
- `SPECIFICATION.md` §3 (roles + app classes), §8.1 (MVP DoD), §9 (D2 client core).
- `04-security-privacy-pki.md` §0.1 (S1–S11 invariants), §1.1 (trust gradient), §1.2 (two auth channels), §1.3 (default-deny realization) — authoritative for the RBAC enforcement model.
- `MASTER_INDEX.md` Volume 1 row (declared scope: "Each persona: jobs-to-be-done, journeys, pain points, success") + Volume 3 row (`svc-identity` as the authoritative permission contract).
- Cross-refs forward: `svc-identity.md`, `svc-policy.md`, `identity-and-enrollment.md`, `03-client-core-and-ui.md`.

Honesty note (§11.4.6): the persona success criteria are tied to the 8-criteria MVP DoD where one exists; the RBAC capability matrix is the product-intent view, with every cell whose grant is *not yet fixed by `svc-identity/svc-policy` marked `UNVERIFIED` and defaulted to the narrower grant. No permission was asserted as final that the authoritative control-plane docs have not yet pinned.*

Constitution bindings: §11.4.44 (revision header), §11.4.6 (UNVERIFIED on un-pinned capabilities, conservative default), §11.4.35 (svc-identity wins on conflict; inherits security invariants), §11.4.104 (participant/identity model alignment), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement).